Implementation and Results of a Feed-Forward Process Controller

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ABSTRACT

The Metropolitan Wastewater Reclamation District (Metro District) commissioned a pilot study in 2011 to test an energy saving process control system at the 80 MGD North Secondary (NSEC) of its Robert W. Hite Treatment Facility. The piloted process control system utilized a feed-forward controller to optimize the dissolved oxygen (DO) set points based on real time influent loading. By linking the DO set points to the loading, the set points dynamically change throughout the day. After using computer software to simulate the predicted savings of utilizing dynamic DO set points, three of twelve trains were converted to utilize the controller and were compared to a baseline set of three trains with constant DO set points. The set of trains utilizing dynamic DO set points required an average of 11% less air. Based on these results, the Metro District has converted the entire NSEC to utilize the process control system.

KEYWORDS: Process Control, Process Optimization, Energy Savings

INTRODUCTION

With continuing increases in energy costs and the requirements of biological nutrient removal (BNR), process control has become one of the most important components of the operations of activated sludge processes.

The energy use of the aeration system for the activated sludge process accounts for approximately sixty percent of the total energy use of the plant (EPA, 1989). A well designed and implemented process control system can save significant amounts of energy and provide a stable BNR effluent.

The purpose for process control at wastewater treatment plants (WWTP) is to respond to disturbances. WWTPs are continuously subject to disturbances caused by the inflow side of the process. The wastewater influent varies in flow, concentration, and composition throughout the day and year. The WWTP must still maintain consistent effluent quality despite these diurnal fluctuations. A process control system with proper instrumentation will minimize the effect of diurnal fluctuations on the effluent quality while achieving treatment goals.

Dynamic dissolved oxygen (DO) set point control has started to become popular with the increased cost of energy and the emergence of better online instrumentation. The basic concept

of dynamic DO set point control is to vary the DO set point with the influent loading to meet an effluent goal. Successful implementation can provide aeration energy savings of 10 to 20% while maintaining the effluent quality. There are two control principles, feedback and feed forward, that are used with dynamic DO set point control.

There have been several case studies (Liu, 2005; Walz, 2009) of facilities using ammonium analyzers to enhance the aeration control system. This paper discusses the implementation and results of a feed forward process controller used in an energy savings project.

BACKGROUND

The Metropolitan Wastewater Reclamation District (Metro District) in Denver, CO treats wastewater from Denver and parts of northern Colorado at its Robert W. Hite Treatment Facility (RWHTF), with an average daily flow of about 140 MGD. The RWHTF currently consists of two distinct secondary treatment plants. The South Secondary (SSEC) is a pure oxygen plant, and the North Secondary (NSEC) is an activated sludge plant. The NSEC is set up as four "quads" with three trains each. The trains are configured as a Modified-Ludzak-Ettinger (MLE) process, with three anoxic zones and one aeration zone in the first pass, and one aeration zone in each of the second and third passes. The NSEC is also equipped with centrate and return activated sludge re-aeration basins (CaRRB), but this process was not the primary focus of the study. The NSEC treats approximately 80 MGD.

In 2009, the Metro District spent over $4.8 million in electrical charges, of which approximately one third was for operating the blowers in the North Secondary. In 2010, the Metro District created the NSEC Energy Optimization Initiative. The initiative was formed to allow operating staff to identify and implement operational changes and technologies to reduce energy consumption. The following projects have been implemented at the RWHTF:

SRT Controller: The Metro District created a solids retention time (SRT) calculator that takes account of the dissolved oxygen, pH, and amount of biomass to calculate the current SRT, minimum SRT and SRT safety factor. The Metro District was able to trend data with effluent ammonia concentrations to find the optimal operating SRT safety factor and the balance of biomass between the secondary and side-stream treatment.

CaRRB Optimization: The CaRRB optimization consisted of installing a pH control system to optimize ammonia removal rates. The control system maintains a pH by controlling the amount of airflow into the CaRRB basins.

Anoxic Zone Mixing: This project evaluated the mixing energy required within the anoxic zones and determined that force of the flow through the serpentine baffle satisfied the mixing requirement. The Metro District turned off all of the mixers in first three anoxic compartments.

Blower Control: In 2010, the Metro District limited the number of blowers in operation at any given time to two to reduce power usage by the aeration system. The blower limit creates DO sags during peak diurnal loading, while not having much impact on the effluent ammonia concentrations.

Dynamic DO set point control: The Metro District's Model User Group (MUG) worked with consultants to conduct model simulations to calculate potential aeration savings at different DO Set Points. BioChem Technology (BioChem) also modeled the process to quantify the potential benefits of using feed-forward dynamic DO set point control. The simulations showed that feed-forward control was able to save 12 to 23% in aeration compared to current operating parameters without affecting the effluent quality. Based upon the simulation results, the Metro District contracted with BioChem to demonstrate the energy savings potential of the feed forward controller.

The controller installed for the demonstration project was the Bioprocess Intelligent Operating System (BIOS) as manufactured by BioChem. The BIOS uses a feed-forward control and optimization algorithm to determine the optimal set-points for DO to achieve treatment goals while minimizing energy consumption. The demonstration started on September 1, 2011 and ended in March of 2012.

For the remainder of the paper the BIOS will be referred to as the feed forward controller or controller.

OBJECTIVE

The objective of the pilot project was to verify that dynamic DO control of the aeration system based on influent loading can reduce aeration requirements and thus aeration energy demand, and therefore reduce total energy costs for the plant. The savings has to be achieved without compromising treatment goals. The success of the controller is measured as the total air saved.

METHODOLOGY

Feed-Forward Process Controller Description

Wastewater treatment plants (WWTP) do not typically operate at the design conditions during most of the day, most of the year, and lifetime of the plant, which provides flexibility to optimize the operations. The aeration system of an activated sludge process is the highest energy consumer system within the treatment train. With the use of DO and ammonia control the energy consumption of the aeration system can be lowered tremendously by taking advantage of the additional treatment capacity during diurnal lows.

The purpose of the DO set-point and ammonia control is to meet effluent ammonia targets while minimizing the amount of oxygen supplied to the process. Minimizing the supplied oxygen reduces the loading on the air blowers, resulting in significant energy and monetary savings.

Residual DO in each aerobic stage of the process can be used to control the specific nitrification rate of the biomass. Increasing the DO concentration increases the nitrification rate by increasing the availability of oxygen to the autotrophic biomass. The increase in the nitrification rate results in a decrease in the effluent ammonia concentration. The converse is also true; lowering the DO concentration will increase effluent ammonia.

Control Theory

The DO set-point controller uses a feed-forward control algorithm to match the DO set-points in the aerobic stages to the variations in diurnal loading. The control algorithm utilizes a biological mathematical model that is customized to the plant and the specific process to calculate an ammonia concentration profile throughout the bioreactor. The model is used in an iterative mode with an optimization algorithm to determine the effect different DO set-points will have on the effluent ammonia concentration. The optimization algorithm ensures the DO set-points calculated are the lowest possible to maintain effluent ammonia targets. Figure 1 is a diagram that visual explains the feed-forward DO set-point controller.

Summary of Measured Values

The feed-forward control algorithm requires only the following inputs:

- Effluent Ammonia (mg/l) (End of Aerobic zone)
- Influent Ammonia (mg/l) (End of Anoxic zone)
- Dissolved Oxygen (mg/l) (one in each aeration zone)
- MLSS (mg/l) within the bioreactors
- Temperature (C˚) within the bioreactor
- Influent flow rate (MGD)
- Return Activated Sludge flow rate (MGD)
- Mixed Liquor Return flow rate (MGD)

At a minimum only one set of ammonium analyzers is required; the controller is capable of calculating the influent ammonia concentrations of multiple trains if the flows are known.

Controlled Values:

The residual ammonia concentration at the end of aeration is used as a control set point for all trains. This effluent ammonia concentration target set point is selected by operators. The ammonia set point can be changed on the user interface at any time.

Manipulated Values:

The DO set points for the individual aerobic stages are used to control the bioreactor effluent ammonia concentration. The controller DO set point calculations are limited to the DO range and maximum step change selected by the operator.

Auto Tuning

At daily intervals, the controller will automatically make minor adjustments to the Specific Ammonia Removal Rate (SARR) if a deviation occurs between the model predictions and the measured effluent ammonia from the day before.

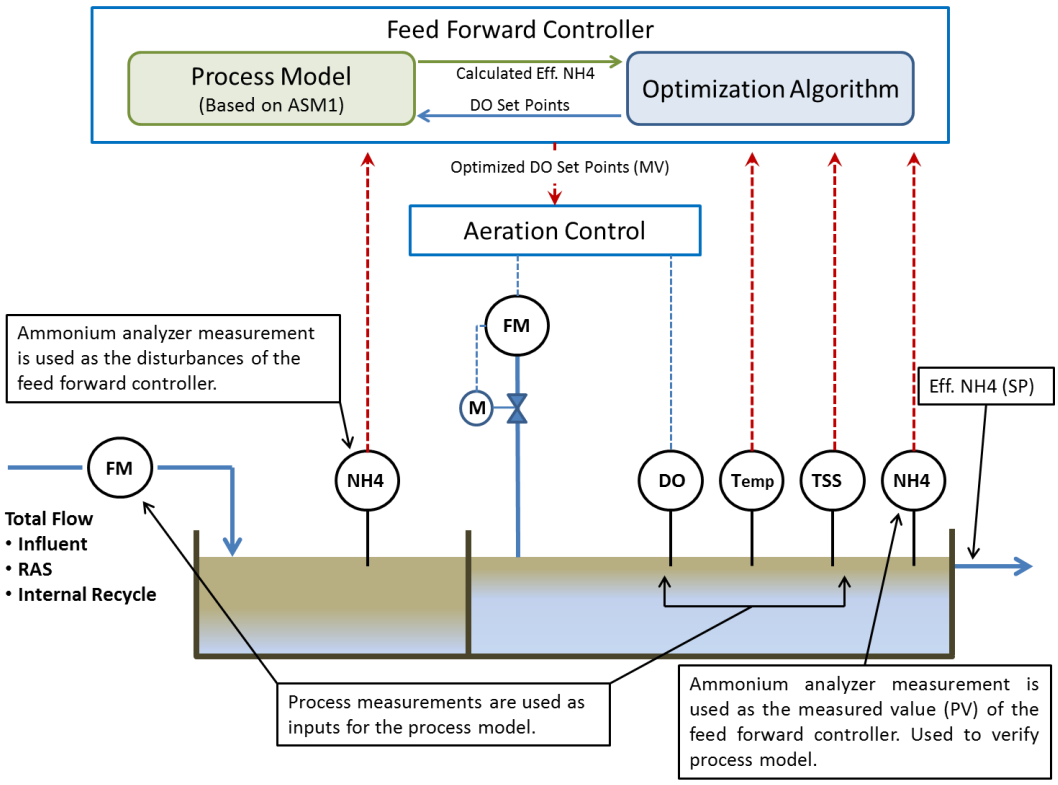

Figure 1. General Dissolved Oxygen and Ammonia Control Diagram

Activated Sludge Model Description

The controller utilizes a mathematical model based upon the International Water Association (IWA) Activated Sludge Model #1 (ASM1) that is customized to the plant specific process. The purpose of the model is to calculate the ammonia concentration within the bioreactor and to work with the process optimization algorithm to calculate the lowest DO set points to achieve the ammonia effluent goal. The model incorporates both the biological kinetics of the activated sludge process and physical characteristics of the specific plant and the treatment process. While it is not a hydraulic model, the model calibration process indirectly accounts for the impact of hydraulic heterogeneities on the treatment performance.

Physical Characteristics

The model only considers the aerobic section of each NSEC train. The aerobic section is modeled as eleven (11) continuously stirred tank reactors (CSTR) in series. The reactor influent (primary effluent), RAS, and mixed liquor recycle (MLR) flows are totalized and used as the flow input to the first aerobic CSTR. The output conditions of the CSTR become the input conditions of the subsequent CSTR. The model's configuration is summarized in Figure 2 below.

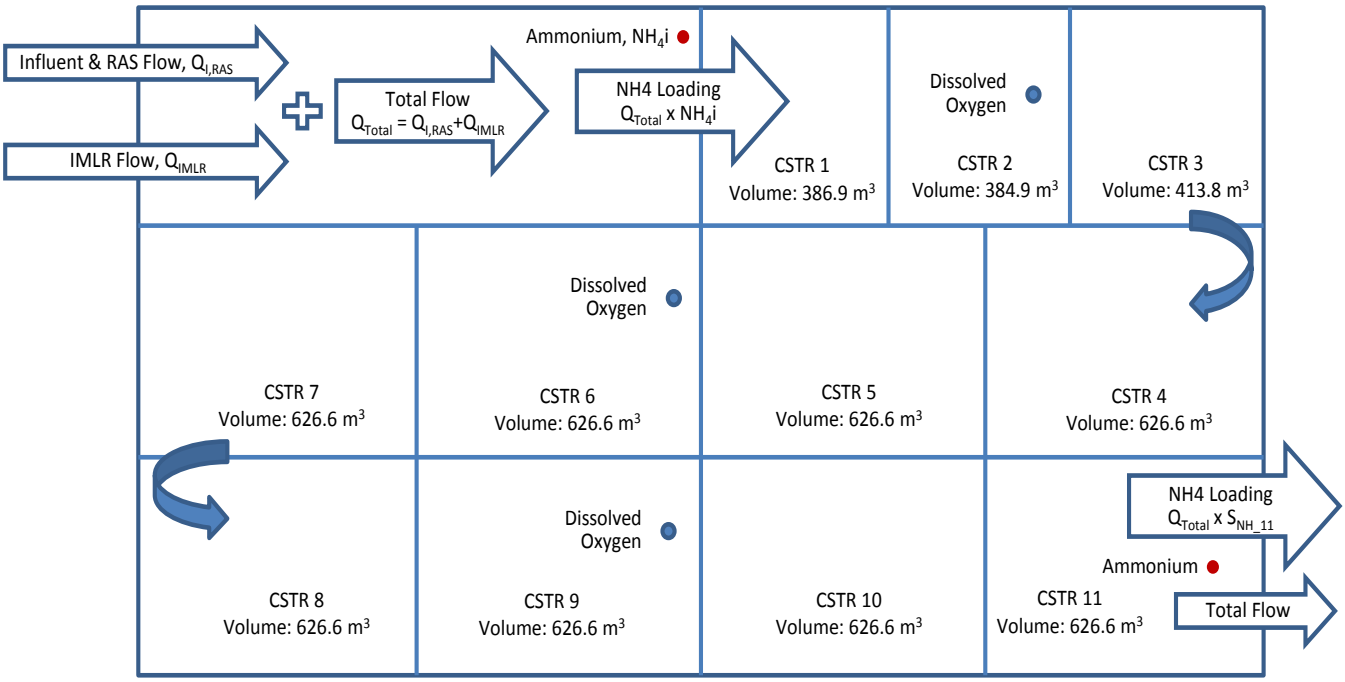

Figure 2. RWHTF NSEC Model Configuration

Biological Kinetics

The purpose of the model is to simulate the concentrations of ammonia within the aerobic reactors, so the model only incorporates process rates from the ASM1 that remove or add ammonia to the process. The IWA model process rates include several parameters that are difficult to measure, making it difficult to tune. Below is the method that BioChem used to modify the original ASM1 process rates.

$$\underbrace{-\left(f_{NH}+\frac{1}{Y_A}\right)}_{\text{Autotrophic Uptake}}\underbrace{\mu_A\cdot\frac{S_{NH}}{K_{NH}+S_{NH}}\cdot\frac{S_{DO}}{K_{DO_A}+S_{DO}}\cdot X_A}_{\text{Nitrification}}+\underbrace{k_a\cdot S_{ND}\cdot X_H}_{\text{Ammonification}}\underbrace{-f_{NH}\cdot\mu_H\cdot\frac{S_S}{K_S+S_S}\cdot\frac{S_{DO}}{K_{DO_H}+S_{DO}}\cdot X_H}_{\text{Heterotrophic Uptake}}$$

$$\underbrace{-\left(\cancel{f_{NH}}+\frac{1}{Y_A}\right)}_{\text{Insignificant}}\underbrace{\mu_A\cdot\frac{S_{NH}}{K_{NH}+S_{NH}}\cdot\frac{S_{DO}}{K_{DO_A}+S_{DO}}\cdot X_A}_{\text{Modified to use measurable variables}}\underbrace{+\cancel{k_a\cdot S_{ND}\cdot X_H}-\cancel{f_{NH}\cdot\mu_H\cdot\frac{S_S}{K_S+S_S}\cdot\frac{S_{DO}}{K_{DO_H}+S_{DO}}\cdot X_H}}_{\text{Incorporated into measured NH}_4\text{ adjustment}}$$

$$\underbrace{SARR}_{\substack{\text{Specific Ammonia Removal Rate}\\ \text{(mg NH4 removed per g of TSS)}}}\cdot\frac{S_{NH}}{K_{NH}+S_{NH}}\cdot\frac{S_{DO}}{K_{DO_A}+S_{DO}}\cdot\underbrace{X_{MLSS}}_{\substack{\text{Mixed Liquor Suspended Solids}\\ \text{(g of TSS)}}}$$

The specific ammonia removal rate is essentially the nitrification process rate with modifications to be able to use measured values instead of the IWA model variables such as growth rate, biomass, and yield for autotrophs. Like the IWA model, the actual ammonia removal rate is dependent upon limiting variables ammonia, and dissolved oxygen.

Feed Forward Controller Program

The program consists of two parts, the user interface and the process control engine.

The user interface is programmed and compiled utilizing the development software Visual Basic by Microsoft. The operational modes and set points can be accessed, implemented, and modified directly through the user interface. Additionally, soft-coded tuning parameters can easily be accessed through the Human Machine Interface (HMI), which allows for on-the-fly control tuning, giving plant operators and engineers the ability to adjust the system as necessary. All data necessary for controller operation is easily visible on the user interface with the train overview screens and historical graphing

The process control engine is where the process model and optimization algorithm calculations are implemented. The mathematical software MatLab was used to develop the process control engine.

Pilot Installation

For the NSEC pilot installation, the controller is programmed to control one quad (Trains 7, 9 and 11). Train 9 is instrumented with ammonium analyzers in the first anoxic zone and at the end of the last aeration basin. The instrumented train is considered the “master train”, and all three trains are controlled based on ammonia measurements in the master train. The controller calculates DO set-points for each of the three aeration zones (Pass A, Pass B, and Pass C) in each of the three trains. A parallel quad (Trains 8, 10 and 12) is instrumented identically to the

controller quad and acts as a baseline for the demonstration. The DO set points for the baseline trains were set by the plant operators. Figure 3 is an aerial view of the NSEC that depicts the feed-forward controller and baseline quads, and Figure 4 is a schematic of the demonstration.

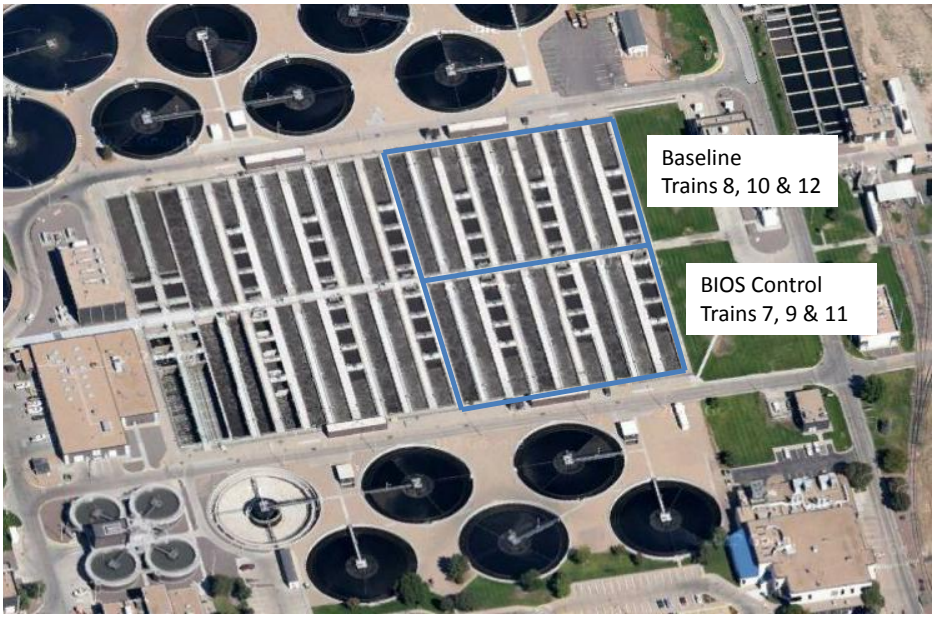

Figure 3. Aerial View of the NSEC at the RWHTF

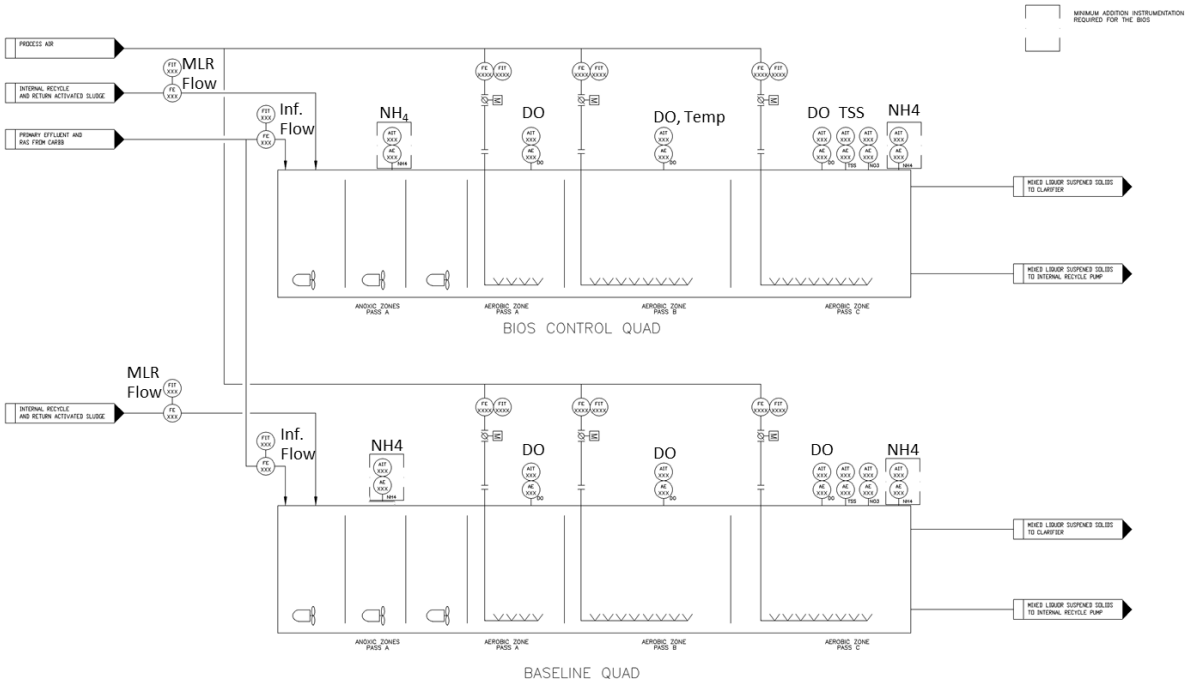

Figure 4. Process Flow Diagram of the Feed-Forward Controller Demonstration

The controller runs on a virtual Windows 7 machine embedded in the existing Metro District server. It communicates with the plant Distributed Control System (DCS) through a Structured Query Language (SQL) Server Express database. The controller retrieves plant operating data

from the database, and transmits DO set-points for each aeration zone to the database for implementation by the plant DCS. For the pilot installation, the controller user interface allows operator input of set-point limits, operating data and manual overrides. The controller was installed in the second half of August, and it began operation on September 1, 2011.

RESULTS

The controller uses a feed-forward control algorithm to provide the required DO set points to meet the variation in diurnal loadings. The aeration energy savings are generated by lowering the DO within the aeration tanks as needed by the process. As DO levels drop, the oxygen transfer efficiency increases, requiring less airflow to meet the same oxygen demand.

A typical week of DO set-points for each pass of Trains 9 (Feed-forward) and 10 (baseline) is shown on Figure 5. In general the DO set-points for Train 9, the feed-forward controlled train, were below the DO set-points for Train 10, the baseline train. The lower DO set-points resulted in decreased airflow usage, as shown on Figure 6. Despite the decreased airflow usage, nitrification performance of the two trains was similar, as shown in Figure 7. The two trains had similar influent ammonia concentrations, and similar effluent ammonia (NH_3-N) and nitrate (NO_3-N) concentrations. This indicates that the lower DO set-points do not have a negative effect on nitrification performance of the secondary treatment.

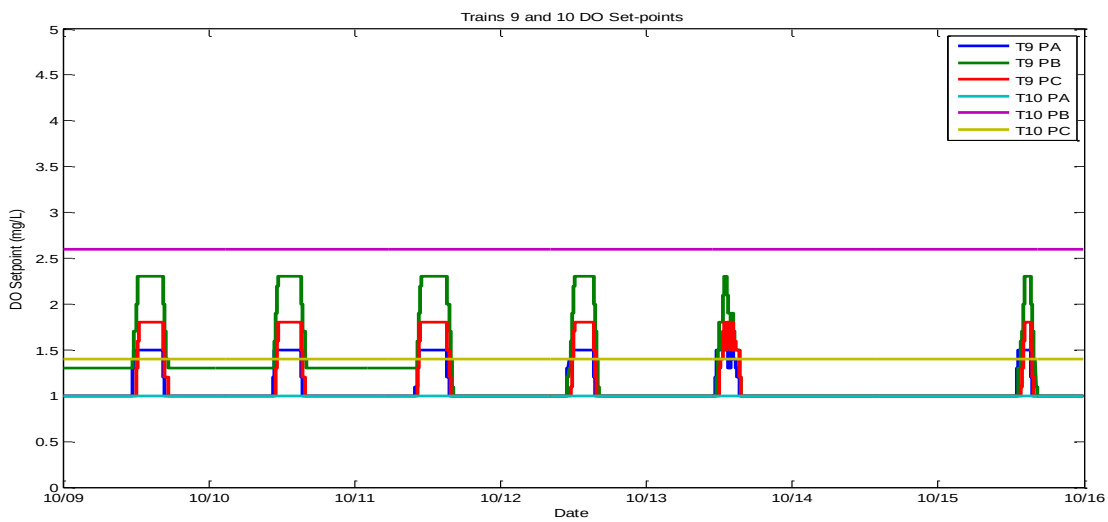

Figure 5. Train 9 and 10 DO Set-Points

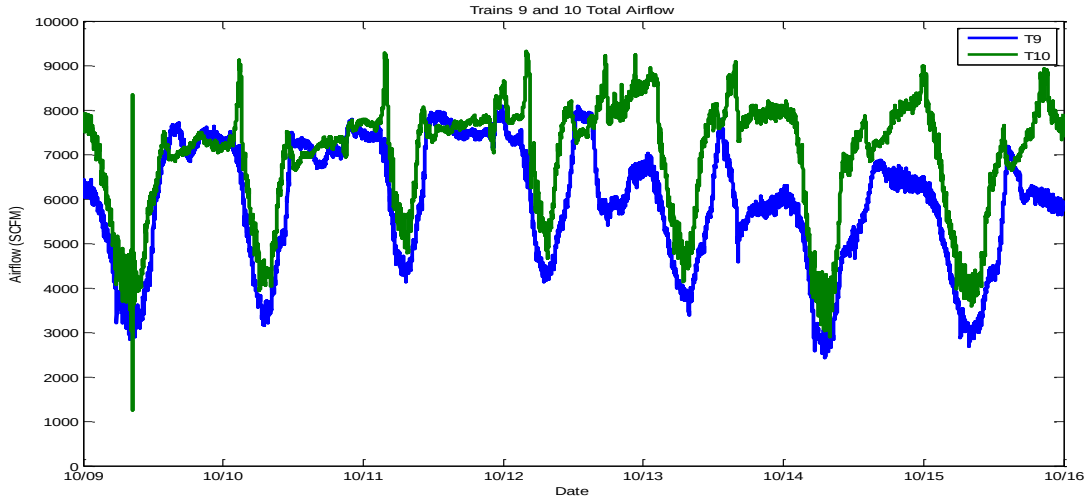

Figure 6. Train 9 and 10 Total Airflows

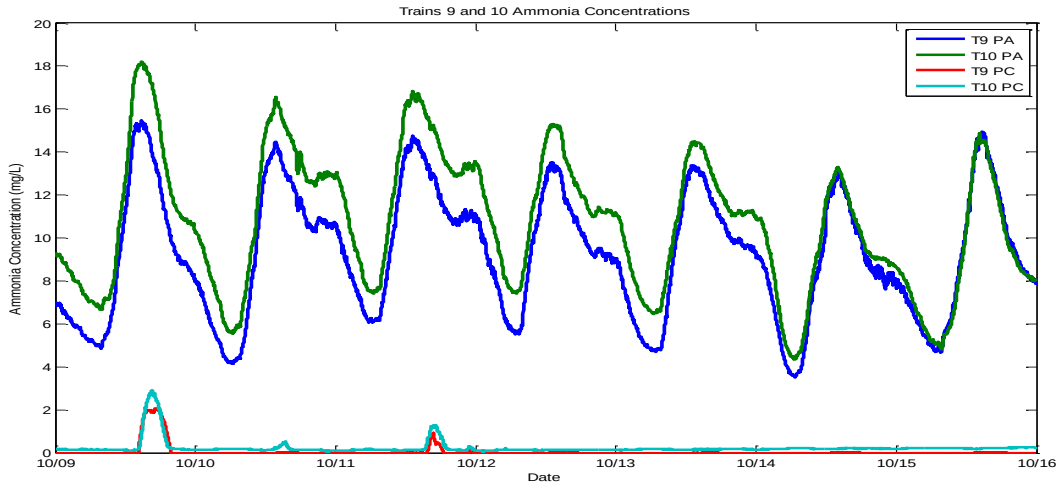

Figure 7. Train 9 and 10 Ammonia

The total airflow to each train was used to compare the feed-forward controlled quad to the baseline quad. The results of the comparisons are summarized on Figure 8. The airflow reduction between the baseline and feed-forward quads was up to 20%. During the six months of the demonstration project the BIOS controlled quad used 11% less air than the baseline quad.

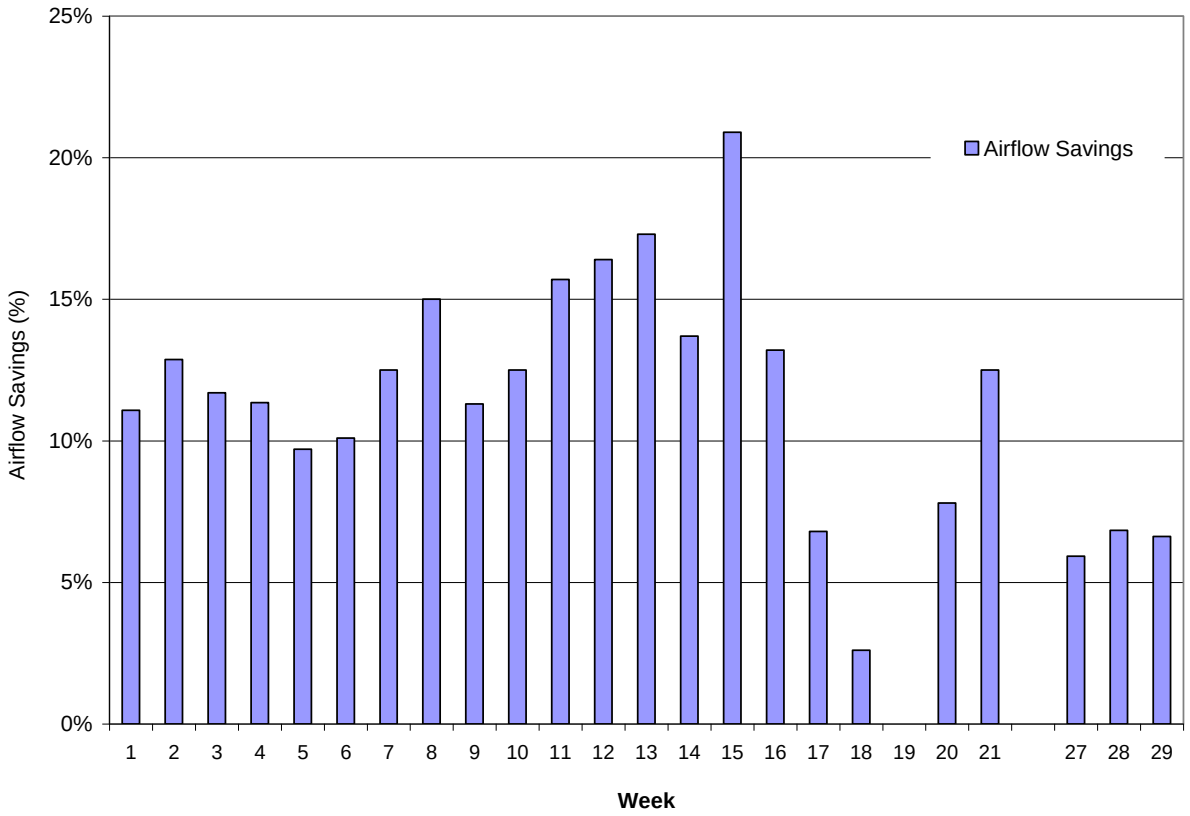

Figure 8. BIOS Aeration Savings

Prior to week 17 the average savings was 13.5%, but during week 17 the operator lowered the DO set-point for the Baseline quad and the other two quads not controlled by the controller. Because the controller DO set points were consistently lower than the baseline while meeting the effluent ammonia limit, operations felt comfortable dropping the DO set points for the rest of the plant to take advantage of the potential savings. As expected, the relative savings generated by the controller dropped, but it was still able to achieve savings of 6%. With further optimization of the selected DO range and effluent goal, the future savings can be increased. Note that the pilot was stopped during weeks 22 to 26 while the DCS system was down for maintenance and upgrades.

The feed-forward controller was part of the Metro District's NSEC Energy Optimization Initiative. Prior to the pilot installation and as part of the Initiative, the Metro District limited the number of blowers in operation to two. This limitation of blowers generated a DO sag during the peak of the day, but reduced the airflow usage by an estimated 10%. During the pilot the feed-forward controller was not affected by the limits on the number of blowers, and only required two blowers independent of the blower restriction. Comparing the DO set-points calculated by the feed-forward controller to the DO set points utilized by the plant before the NSEC Energy Optimization Initiative was started shows a theoretical savings of 22-30% using the feed-forward controller.

To place a monetary value to the observed savings, the observed airflow differentials were entered into the following equation, which includes the Metro District specific power cost in 2010 ($0.075/kWh) and conversion factor for airflow to power (0.000423 kilowatt hour per standard cubic foot [kWh/scf]):

$$Daily\ Aeration\ Cost\ (\$/d) = Airflow\ \frac{scf}{min}\ x\ 0.000423\ \frac{KWh}{scf}\ x\ 1440\ \frac{min}{day}\ x\ \frac{\$0.075}{kWh}$$

The estimated aeration savings during the demonstration are approximately **$18,635**. With a full installation of the BIOS, the estimated yearly savings would be **$134,026** at the 11% observed average savings.

DISCUSSION

The following are discussion points that were deemed import to mention in the paper.

Simulated Versus Actual Savings

Prior to the pilot project, the Metro District's MUG (in conjunction with Black & Veatch) and BioChem separately applied commercial simulation software to quantify the potential benefits of lower the DO set points using dynamic DO set-point control. A model was created using influent characteristics, and operational and physical parameters of the NSEC facility. The projected simulated savings versus actual is summarized in Table 1.

Table 1. Summary of the Simulated Savings versus Actual Results from the Pilot

Temperature	Eff. NH_4	Aeration Savings	Nitrate Reduction
C	mg/l	%	%
B&V Simulation			
15	2.5	16	29
18	1.9	22	28
23	1.1	17	12
BioChem Simulation			
15	0.6	13	15
20	0.5	16	14
24	0.4	23	21
Pilot Project Results			
18	1-5	15-20	7

When the feed-forward controller was able to utilize the same lower DO range as the simulations the actual aeration savings were close to what was projected by the simulations. There was some aeration savings difference between the two simulations, but that was mostly due to the differences in the simulation inputs.

NSEC Aeration System

To fully utilize the benefits of the feed-forward controller, it is essential to have an aeration system that is capable of controlling the airflow to meet the dynamic DO set points calculated by the controller. In its default setting the controller limits the DO set point change to 0.3 mg/l per calculation cycle of 15 minutes. The NSEC aeration control system shows an acceptable level of performance for control. Aeration control over a three day period in Train 9 Pass B is shown on Figure 9.

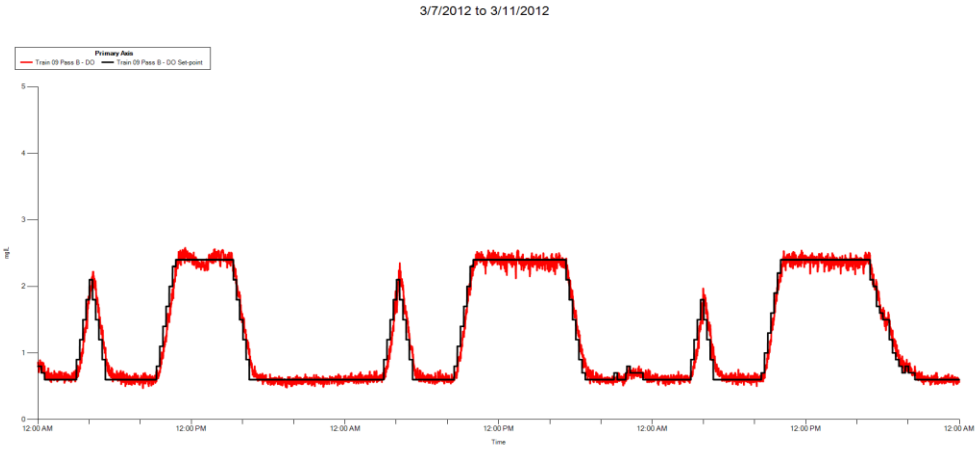

Figure 9. DO Set-Point Tracking for Train 9 Pass B

Ammonium Analyzers

With any control system the control is only as good as the instrumentation that is used. For the feed-forward control system there are several instruments required: flow, DO, temperature, MLSS, and ammonium analyzers. Most of the instrumentation is well established within the wastewater community and requires limited maintenance. In recent years ammonium analyzers have made great strides in regards to robustness, but they still require frequent maintenance and calibration. The Metro District checks the analyzers once a week with grab samples and adjusts the analyzer calibrations curves as needed.

The Metro District provided and installed four (4) Endress & Hauser (E&H) ammonium analyzers based upon BioChem's recommendations. Overall, the ammonium analyzers worked well. Train 9 and 10 analyzer readings have been closely tracking each other, with only few periods of questionable readings. Figure 10 shows ammonium readings during a typical day.

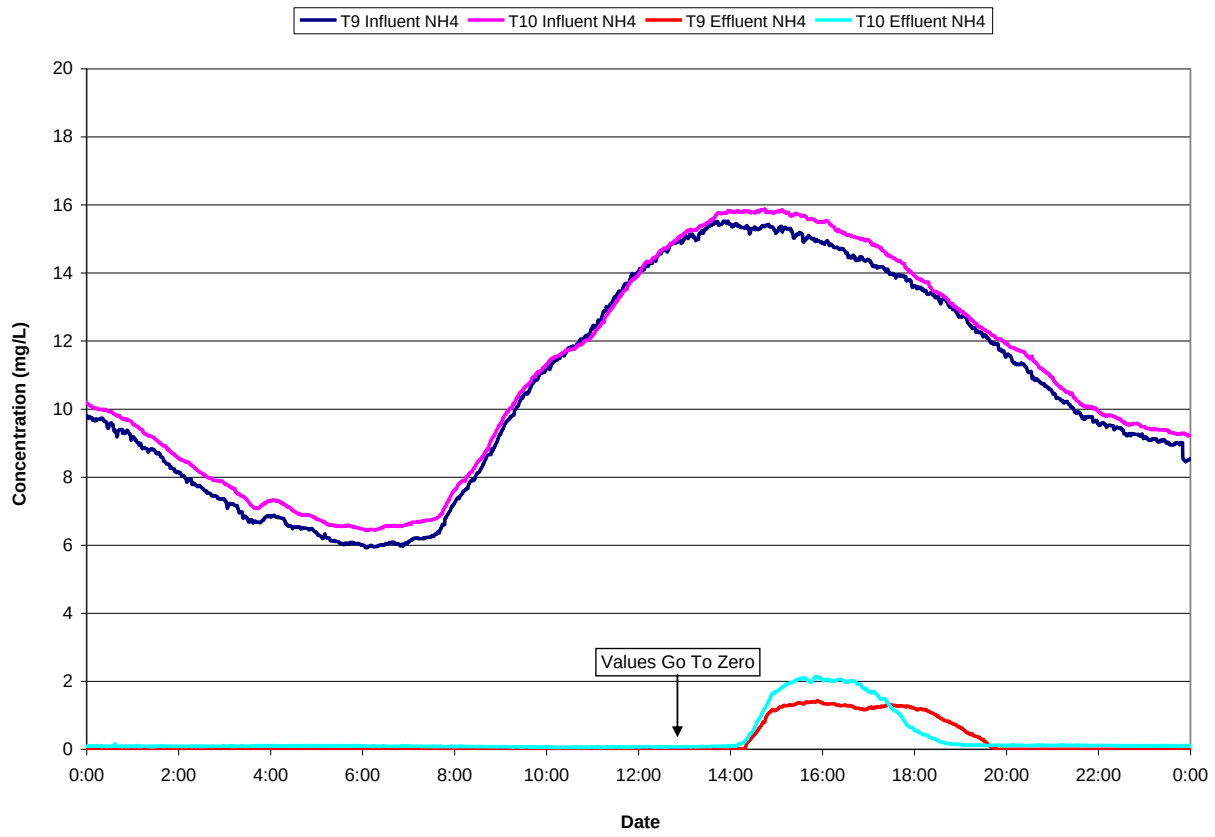

Figure 10. Train 9 and 10 Ammonium Analyzer Measurements

To ensure that the Metro District has continuous aeration savings, uninterrupted by ammonium analyzer faults, BioChem developed a software module to monitor and back up the ammonium analyzers located in the anoxic zone. The software module tracks influent ammonium loading average and standard deviation for each hour of the day.

The use of average data and standard deviation is a common way to detect data outliers. Assuming that the data follows a normal distribution, 99.7% of the data will fall within three standard deviations of the average. Data outside of three deviations will be considered an outlier. Data that are considered an outlier will be excluded from future calculations of average and standard deviation. Figure 11 is a chart of Train 9 influent ammonium analyzer statistical analysis.

The controller also checks the current ammonia loading measurement for outliers. If the data point is considered an outlier the controller will notify operations with a warning. The controller will continue to use the measured data until operations decides the analyzer is in fault and that the backup method should be used. The backup method will use an operator selected safety factor multiplier of the historical ammonium loading to calculate the concentration within the anoxic tank.

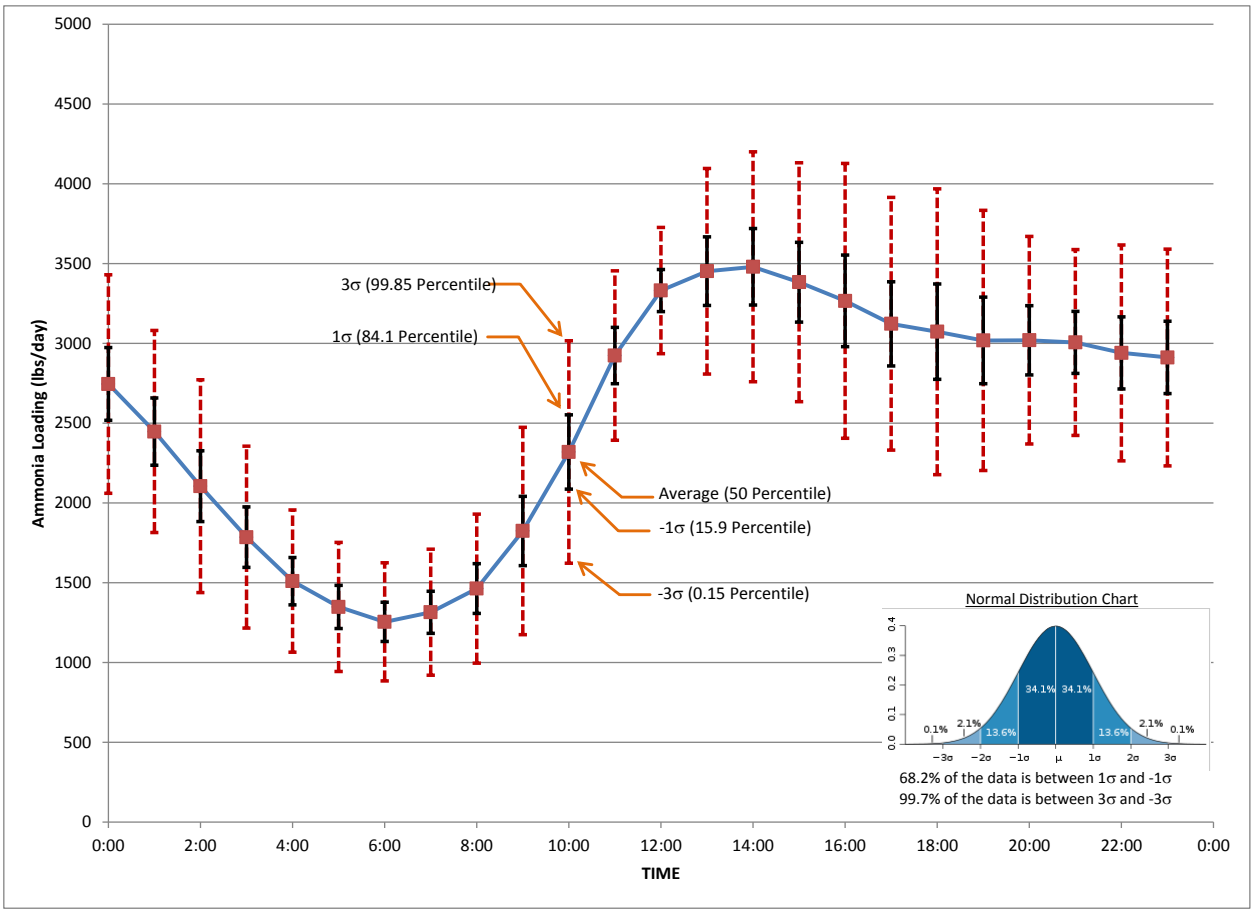

Figure 11. Train 9 Influent Ammonia Analyzer Statistical Analysis

Potential Denitrification Benefits

Using dynamic DO set-point control can lower effluent nitrate, assuming an ideal fraction of readily biodegradable COD is available. Overall denitrifcation is increased because the use of lower DO set-points recycles less oxygen to the anoxic zone, and the potential of simultaneous nitrification denitrification is higher. Figure 12 is an example of when SND was seen within the feed forward quad. The figure shows when the DO are high the effluent nitrate from the baseline and feed forward trains are similar, but when the feed-forward train drops the DO the nitrate drops below the baseline nitrate.

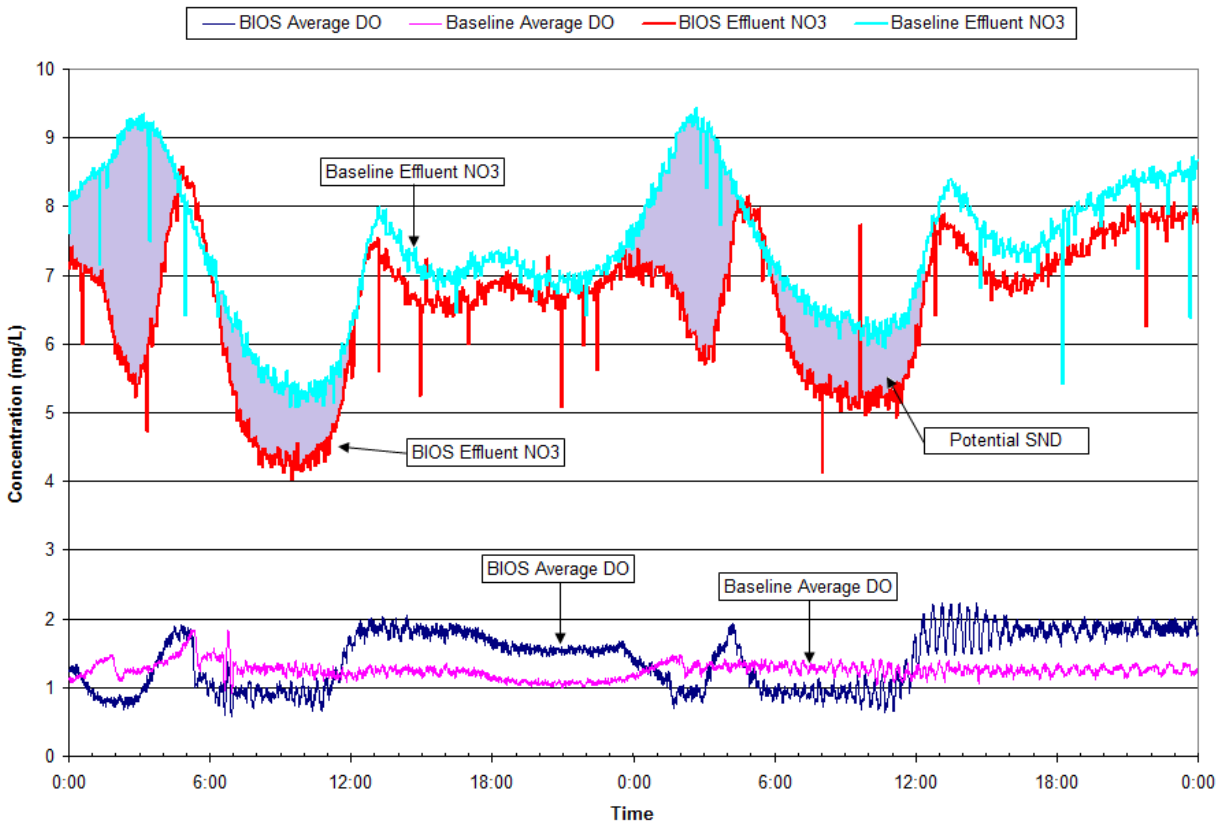

Figure 12. Potential SND within the BIOS Quad

Dissolved Oxygen Profile

The original control model assumed that that the DO profile is constant across an entire pass due to the inability to measure DO at more than one point in a pass. However, when the DO set-point drops to 0.5 mg/L, it is possible that portions of Pass B have very low to no measurable DO present during higher loadings, leading to a severe decline in the nitrification rate that was not accounted for in the model. The presence of very low to no DO zones would cause the large measured effluent ammonia spikes.

To improve the controller effluent prediction, a DO profile was applied to the model, which would more realistically capture the aeration performance of the pass. The profile used is shown in Table 4.1 and was based upon actual DO profiles measured by the Metro District. The model assumes there are four aeration zones per pass, except Pass A where there are three zones.

The use of the new DO profile was simulated using actual data collected from the controller, and the simulation showed improved predictions of effluent ammonia. The results of the simulation are shown on Figure 13.

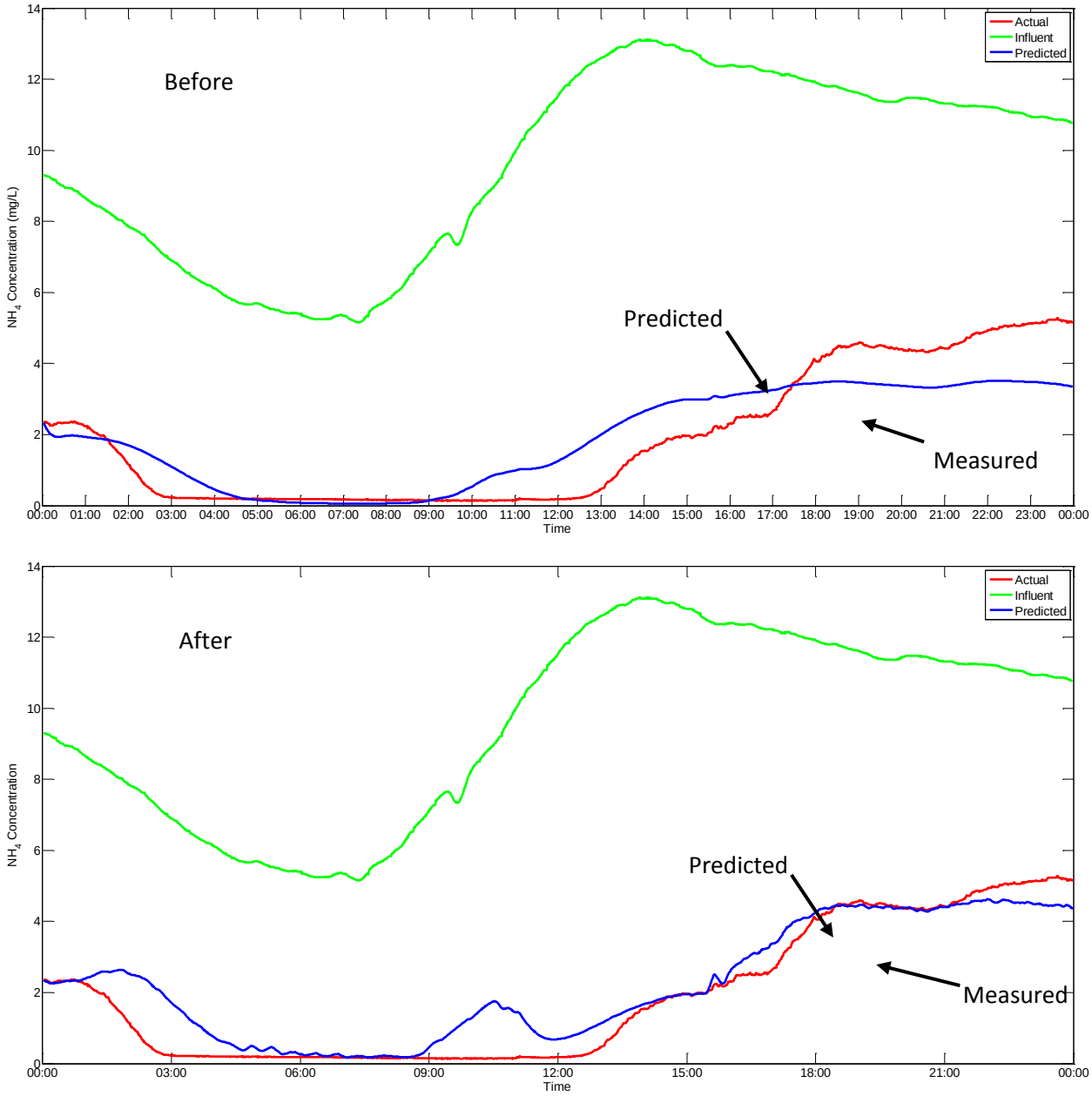

Figure 13. BIOS Effluent Prediction Before and After DO Profile Modification

Ammonia Effluent Prediction

The original control model assumed the process is operated with a constant SRT safety factor, where the fraction of nitrifying biomass has small variations and is mostly dependent on the ammonia loading, so the ammonia removal rate variation is small. As part of the NSEC Energy Optimization Initiative, operations found that the optimal SRT Safety factor is 0.90. Activated sludge operating at a lower SRT safety factor has variations in the fraction of the nitrifying biomass and ammonia removal rates. Varying ammonia removal rates lead to under- or over-estimating the effluent ammonia concentration.

To ensure that the ammonia effluent prediction is close to the measured value, the controller auto tunes the Specific Ammonia Removal Rate (SARR). At daily intervals, the controller automatically makes minor adjustments to the SARR if a deviation occurs between the model predictions and the measured effluent ammonia from the day before. Figure 14 is three days of the auto tuning.

The rate by which the auto tuning adjusts the SARR is limited to a selected percent and it must stay within a range. The limitation prevents an un-calibrated effluent ammonium analyzer from driving the model in the wrong direction. Effluent measurements from ISE ammonium analyzers need to be carefully examined. E&H ISE ammonium analyzers have an accuracy of 0.2 mg/l (E&H 2011), and ISE calibration coefficients are not applicable at the non-linear response seen at lower concentrations (Fry, 2001), so the effluent readings are susceptible to incorrect readings.

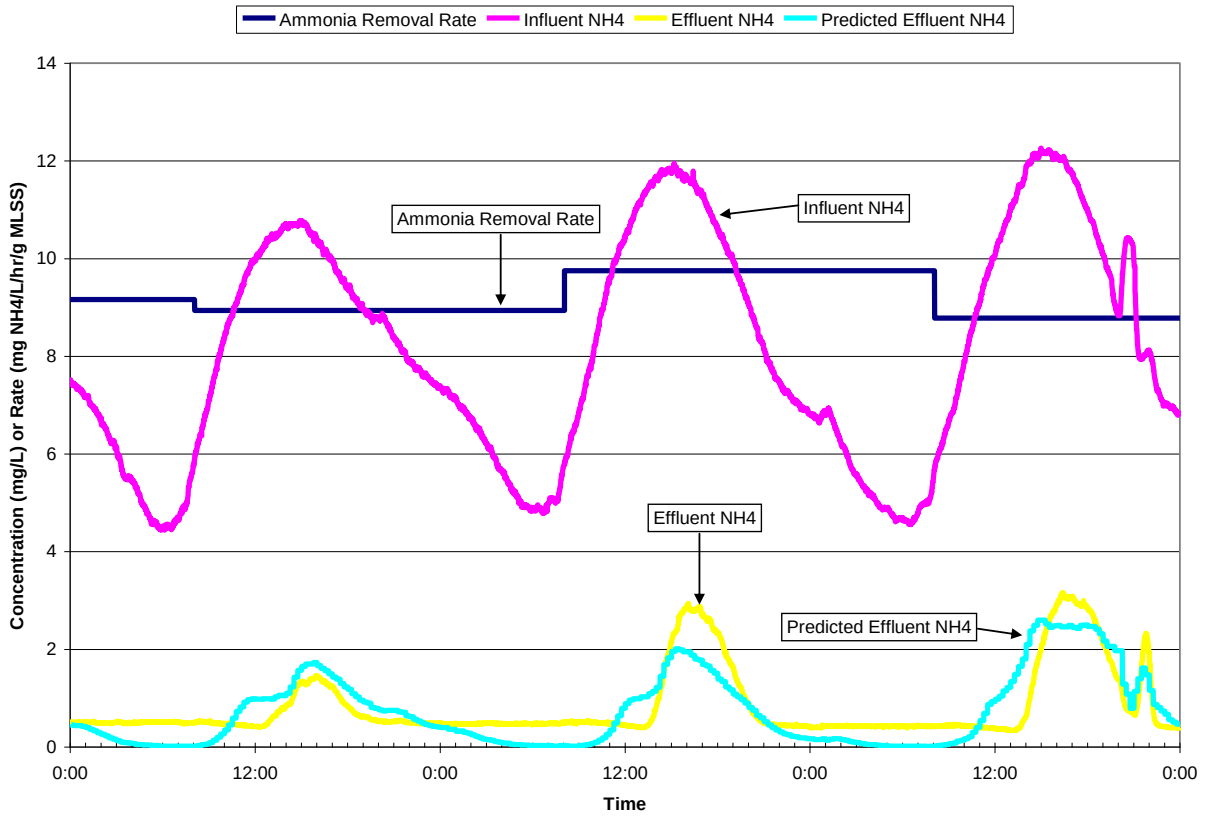

Figure 14. Feed-Forward Controller Auto Tuning

CONCLUSION

Overall the feed-forward controller (BIOS) pilot project was a success. The objective of the pilot project was to verify that dynamic dissolved oxygen control of the aeration system based on feed-forward control. Over the 6 months of the pilot, the feed-forward controlled quad used 12% less air than the baseline quad. With further optimization of the selected DO range and effluent goal, the savings can be increased. The estimated aeration savings during the demonstration, as a

direct result of reduced airflow requirements, is approximately **$18,635**. With a full installation of the BIOS, the estimated yearly savings is **$134,026.**

The feed-forward dynamic DO set-point control pilot convinced the Metro District to proceed with a full installation. The full installation required some customization of the software, but no additional instrumentation needed to be installed for control purposes. The following is the summary of the finalized control system highlights:

- Calculates DO set points for each of the three (3) aeration zones in the twelve (12) process trains, for a total of thirty-six (36) individual calculated DO set-points.

- Two (2) sets of ammonium analyzers are used for control purposes; an additional set has been installed for analyzer measurement comparisons by the DCS.

- The controller collects historical ammonia loading measurements, and calculates hourly averages and standard deviations. The calculations are used to monitor the ammonium analyzers, and as backup if the analyzers are deemed in fault, to ensure continuous operation of the controller.

- Each train has individual control settings from ammonia effluent set point to the allowable DO set point ranges. This allows for future controlled ammonia "bleed-through" to support disinfection by formation of chloramines.

- Effluent ammonium readings are used to modify ammonia removal rates within the model. This will ensure that the effluent ammonia is as close to the effluent target as possible.

- The calculated removal rate values will be stored with the SRT safety factors to create a historical correlation. With a full set of correlation data the BIOS will be able to detect possible process upsets.

CREDITS

The authors acknowledge the contributions and support of the staff of the Metro District Robert W. Hite Treatment Facility. Specifically Kevin Feeley, Perry Holland, John Kuosman, Craig Barnes, Bill Wendt, Dan Brown, Liam Cavanaugh, Mike Elfert, Dave Lozano, David Hudgins, and Fred Hayes were all instrumental in ensuring the success of this work.

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